Layer- and regime-dependent interpretability in concept bottleneck models for single-cell transcriptomics

Sally Yepes
sallyepes233@gmail.com

Abstract

Concept-bottleneck models provide a structured approach to interpretable machine learning in biology by aligning intermediate representations with predefined biological concepts, such as transcription factor regulatory programs. However, it remains unclear whether commonly used indicators of interpretability—such as activation selectivity, structured embeddings, or agreement with known targets—reflect a unified notion of mechanistic understanding.

Here, we introduce a multi-layer framework that decomposes interpretability into three complementary components: concept selectivity, concept-space geometry, and gene-level mechanistic support. These components are evaluated on the same learned representation but capture distinct aspects of model behavior. We apply this framework to concept-bottleneck models trained on single-cell transcriptomic datasets spanning multiple biological regimes, including stimulus-driven responses, a more directly regulated interferon setting, indirect perturbations, and heterogeneous perturbations.

Across all datasets, structured organization in concept space is consistently observed. However, gene-level support varies substantially by regime: it is partial in stimulus-driven data, strongest in the interferon setting, absent in indirect perturbations, and heterogeneous in large-scale perturbation datasets. Notably, structured representations frequently arise without corresponding agreement with known regulatory targets, demonstrating a decoupling between representation-level organization and mechanistic validity.

These findings establish that interpretability in concept-bottleneck models is inherently multi-dimensional and regime-dependent. More broadly, they show that representation structure alone is insufficient to support mechanistic conclusions and that interpretability must be evaluated using multiple complementary criteria.

1. Introduction

Interpretable machine learning has become an important objective in computational biology, where models are increasingly used to analyze high-dimensional gene expression data and generate biologically meaningful insights. Concept-bottleneck models and biologically structured neural networks offer a promising approach by organizing intermediate representations around predefined biological concepts, such as transcription factor (TF) regulatory programs (Fortelny and Bock, 2020). In these models, interpretability is enforced through model structure rather than inferred post hoc.

Despite this appeal, it remains unclear to what extent concept-based representations provide mechanistically meaningful interpretations. In practice, interpretability is often assessed using multiple criteria, including activation selectivity, structure in concept space, and agreement with known biological annotations. These criteria are frequently treated as interchangeable, even though they probe different aspects of model behavior.

A key challenge is that interpretability is not a single property but a combination of distinct characteristics that may not align. A model may produce highly selective concept activations while failing to recover expected downstream targets, or exhibit structured organization in concept space without correspondence to known regulatory mechanisms. This distinction is consistent with prior work in machine learning interpretability, where different methods capture complementary aspects of model behavior rather than a single unified notion of explanation (Lundberg and Lee, 2017; Zeiler and Fergus, 2014).

Here, we introduce a multi-layer framework that separates interpretability into three components evaluated on the same learned representation: (i) concept selectivity, (ii) concept-space geometry, and (iii) gene-level mechanistic support. This separation enables systematic analysis of when different aspects of interpretability converge or diverge.

We apply this framework to concept-bottleneck models across multiple single-cell datasets representing distinct biological regimes, including stimulus-driven peripheral blood mononuclear cell (PBMC) responses (Kartha et al., 2022), a more directly regulated interferon (IFN) setting (Kang et al., 2018), indirect perturbations (Norman et al., 2019), and heterogeneous perturbations (Replogle et al., 2022). These regimes vary in how directly transcriptional responses map to known regulatory programs.

Across all datasets, structured organization in concept space is consistently observed. However, gene-level support varies substantially: it is partial in PBMC, strongest in the IFN setting, absent in Norman, and heterogeneous in Replogle without clear dependence on regulator class. These results demonstrate that representation structure does not guarantee mechanistic validity.

Together, our findings show that interpretability in concept-bottleneck models is multi-dimensional and regime-dependent. They highlight the need to evaluate interpretability using multiple complementary criteria and to distinguish explicitly between representation-level organization and mechanistic correspondence.

2. Framework: multi-layered interpretability

Interpretability in concept-bottleneck models is often treated as a single property of the learned representation, yet different analyses probe distinct aspects of model behavior and may not yield consistent conclusions. We therefore introduce a multi-layered framework that decomposes interpretability into complementary components evaluated on the same concept representation.

The concept representation is a structured intermediate layer corresponding to predefined transcription factor (TF) programs rather than an unconstrained latent embedding, enabling interpretability at the level of biological concepts (Fortelny and Bock, 2020). We define three layers of interpretability: concept selectivity, concept-space geometry, and gene-level mechanistic support, all derived from the same concept activations but capturing different properties.

Concept selectivity measures how strongly activity is concentrated in individual concepts. High selectivity indicates dominance of a small number of concepts, while low selectivity reflects distributed activation. This provides a local view of model behavior but does not establish biological validity.

Concept-space geometry characterizes the global organization of samples or perturbations in the space of concept activations. Structured geometry reflects consistent relationships among conditions but does not imply correspondence with known regulatory mechanisms.

Gene-level mechanistic support evaluates alignment between concept activations and expected downstream transcriptional effects by comparing attributed genes with curated target sets. High overlap suggests agreement with known regulatory programs, while low overlap indicates divergence and depends on annotation completeness (Garcia-Alonso et al., 2019).

These layers are complementary but independent: high selectivity or structured geometry does not guarantee gene-level support, and gene-level support may occur without global structure. This separation is consistent with prior work showing that interpretability methods capture distinct aspects of model behavior rather than a unified notion of explanation (Lundberg and Lee, 2017; Zeiler and Fergus, 2014). Figure 1 illustrates this framework, where gene expression inputs are mapped to TF concepts through a concept bottleneck and interpretability is evaluated across the three layers as independent lenses. This framework emphasizes that interpretability should not be inferred from a single metric but evaluated across multiple dimensions, providing a basis for analyzing regime-dependent behavior across datasets.

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3. Results

3.1 Evaluating interpretability as a multi-layer property

We evaluate interpretability in concept-bottleneck models using three complementary analyses applied to the same concept representation: concept selectivity, concept-space geometry, and gene-level mechanistic support. Because the intermediate layer is explicitly constrained to transcription factor (TF) programs, these analyses probe interpretability directly at the level of biologically defined regulatory concepts rather than unconstrained latent features (Fortelny and Bock, 2020).

Concept selectivity quantifies how strongly activity is concentrated in a small number of concepts for a given condition or perturbation. Concept-space geometry captures the global organization of samples or perturbations in the space of concept activations, including similarity structure and low-dimensional organization. Gene-level mechanistic support evaluates agreement between model-derived gene attributions and expected downstream targets. These analyses provide complementary views of interpretability and do not imply a hierarchy (Lundberg and Lee, 2017; Zeiler and Fergus, 2014).

3.2 PBMC stimulation produces stimulus-specific concept activation patterns

We first evaluated MM-KPNN in a PBMC stimulation dataset, where transcriptional variation is driven by relatively well-characterized immune signaling programs (Kartha et al., 2022). This setting provides a favorable regime for testing whether concept activations align with known biology.

Across stimulation conditions, the model produces distinct concept activation profiles corresponding to known regulatory responses. Interferon stimulation is associated with strong induction of interferon-response concepts, including STAT2, IRF9, and CIITA, while lipopolysaccharide stimulation activates NFκB-associated concepts and PMA induces mixed AP-1–related activity. These patterns are visible in the concept activation heatmap and in activation shifts relative to control (Figure 2A–B).

Concept activations are selective rather than diffuse, indicating that the model concentrates activity within stimulus-relevant regulatory programs. This dataset therefore defines a regime in which concept-level responses are biologically plausible and interpretable at the level of activation patterns.

3.3 PBMC reveals partial alignment between concept activity and gene-level support

Although PBMC concept activations are biologically coherent, gene-level support is not uniform across all stimuli and concepts. Figure 2C shows the relationship between activation shift and overlap with expected downstream targets. Some concepts, such as STAT2 under interferon stimulation, exhibit both strong activation and strong overlap, indicating alignment between concept activity and expected regulatory effects. Other concepts show weaker or inconsistent support despite clear activation.

This pattern indicates that the PBMC regime is not uniformly mechanistic. Instead, it contains instances of strong alignment alongside cases of weaker agreement. Even in a favorable, stimulus-driven setting, strong concept activation does not universally translate into strong downstream support. Mechanistic interpretation is therefore feasible but conditional and subset-dependent.

3.4 IFN stimulation defines an upper-bound regime for interpretability alignment

To examine model behavior in a more directly regulated setting, we evaluated MM-KPNN on an interferon (IFN) dataset (Kang et al., 2018). In this regime, transcriptional responses are dominated by well-characterized interferon signaling programs.

This setting approximates an upper-bound regime in which architectural constraints align closely with the dominant biological signal. Concept activations are highly selective and concentrated in interferon-associated programs, and perturbations form well-separated patterns in concept space (Figure 2A–B). Gene-level support is consistently above a shuffled baseline, indicating measurable enrichment of relevant regulatory targets.

However, overlap remains incomplete, and not all strongly activated concepts exhibit high agreement with expected targets. Even in this favorable regime, mechanistic correspondence is partial rather than complete.

3.5 Norman perturbations produce structured concept-space geometry

We next evaluated MM-KPNN in the Norman Perturb-seq dataset, which introduces targeted perturbations that induce complex and often indirect transcriptional responses (Norman et al., 2019).

Low-dimensional projections of perturbation-induced concept responses reveal organized structure in concept space, with perturbations forming non-random patterns in PCA embeddings (Figure 3A). This indicates that the model captures stable relationships among perturbations at the representation level, despite the indirect nature of the underlying effects.

The Norman dataset therefore provides a clear example of structured concept-space organization in a regime dominated by distributed and indirect effects.

3.6 Norman shows no measurable gene-level support despite structured responses

In contrast to the structured geometry, gene-level support in the Norman dataset is weak. Observed overlap with expected targets is indistinguishable from a shuffled baseline (Figure

3B), indicating no measurable enrichment. Concept selectivity also does not predict gene-level support (Figure 3C).

This defines a regime in which architectural interpretability is present—concepts are well-defined and responses are structured—but mechanistic support is absent. Structured representations can therefore arise independently of gene-level interpretability, demonstrating a clear decoupling between representation structure and mechanistic grounding (Lundberg and Lee, 2017).

3.7 Replogle perturbations exhibit heterogeneous interpretability without class-level separation

We next analyzed the Replogle CRISPR perturbation dataset, which spans transcription factors, chromatin regulators, and other regulatory proteins (Replogle et al., 2022).

Gene-level support varies substantially within each regulator class (Figure 4A), and concept selectivity spans a wide range across perturbations (Figure 4B). At the perturbation level, strong and weak cases appear across all classes, indicating that variability is driven by individual perturbations rather than class-level properties (Figure 4C).

This defines a regime of heterogeneous interpretability. Concept-space representations remain structured, but gene-level support is uneven and context-dependent. Interpretability varies at the level of individual perturbations and cannot be inferred from regulator class.

3.8 Cross-dataset synthesis reveals regime-dependent interpretability

We compared PBMC, IFN, Norman, and Replogle to identify which aspects of interpretability are preserved across datasets and which depend on biological context.

Across all settings, structured organization in concept space is consistently observed, reflecting a robust property of the architecture. In contrast, gene-level support differs substantially by regime: partial in PBMC, strongest in IFN, absent in Norman, and heterogeneous in Replogle (Figure 5).

Together, these results demonstrate that interpretability in concept-bottleneck models is both layer-dependent and regime-dependent. Structured representations are stable across datasets, but mechanistic support depends on the underlying biological context.

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4. Discussion

This study evaluates interpretability in concept-bottleneck models across multiple biological regimes and shows that interpretability is not a single property of a model, but a multi-dimensional and context-dependent phenomenon. By separating concept selectivity, concept-space geometry, and gene-level mechanistic support, we demonstrate that different aspects of interpretability can vary independently.

A central finding is that structured concept-space representations are consistently observed across all datasets, yet gene-level support is not. Perturbations and stimuli organize into coherent patterns in concept space, indicating that the model captures stable relationships at the representation level. However, the degree to which these patterns correspond to expected regulatory targets varies substantially. This dissociation shows that representation structure alone is insufficient to establish mechanistic interpretability.

An important aspect of this work is that interpretability is enforced by design through the model architecture. The masked gene-to-concept mapping constrains intermediate representations to predefined transcription factor programs, allowing interpretability to be evaluated directly at the level of biological concepts (Fortelny and Bock, 2020). At the same time, the results demonstrate that architectural interpretability does not guarantee mechanistic validity. Even when the representation is biologically grounded by construction, agreement with downstream targets depends on the dataset and regulatory context.

Across datasets, distinct interpretability regimes emerge. In the PBMC dataset, concept activations align with known biology and gene-level support is observed for a subset of responses, particularly in interferon-associated programs, but this alignment is not uniform. In the IFN-focused setting, all interpretability layers are strengthened, defining an upper-bound regime in which architectural constraints closely match the dominant biological signal; however, gene-level recovery remains incomplete. In contrast, the Norman dataset reveals a regime in which structured concept-space organization emerges without measurable gene-level support, demonstrating a clear decoupling between representation structure and mechanistic grounding. The Replogle dataset further shows that interpretability can be heterogeneous and perturbation-specific, with gene-level support varying across perturbations and not following a simple regulator class hierarchy.

These observations have important implications for the evaluation of interpretable models in biology. Common indicators such as high concept selectivity or well-structured embeddings reflect meaningful organization within the model but are not sufficient to establish mechanistic validity. Interpretability should therefore be assessed using multiple complementary criteria rather than inferred from a single property of the representation (Lundberg and Lee, 2017; Zeiler and Fergus, 2014).

More broadly, this work highlights the importance of distinguishing between representation-level organization and mechanistic correspondence. Concept-bottleneck models can produce coherent and biologically meaningful representations, but these representations do not uniformly recover known regulatory effects. This distinction is consistent with prior work in machine learning interpretability, where different explanation methods capture complementary aspects of model behavior rather than a unified notion of explanation (Lundberg and Lee, 2017; Zeiler and Fergus, 2014).

Several limitations should be considered. First, gene-level support is evaluated using overlap with curated target sets, which may be incomplete or context-dependent; absence of overlap does not necessarily imply absence of biological relevance. Second, concept definitions are constrained by the underlying regulatory resources and model architecture, which may influence both selectivity and attribution. Third, comparisons across datasets are interpreted qualitatively rather than as directly normalized quantitative measures, reflecting differences in experimental design and perturbation regimes.

Despite these limitations, the framework introduced here provides a systematic approach for evaluating interpretability in concept-bottleneck models across diverse biological settings. By explicitly separating complementary aspects of interpretability and analyzing their behavior across regimes, we show that interpretability is inherently multi-dimensional, regime-dependent, and not fully determined by model structure alone.

Key takeaways

1. Interpretability is multi-dimensional: selectivity, representation structure, and gene-level support capture distinct aspects of model behavior and should not be conflated.
2. Structure does not imply mechanism: coherent organization in concept space does not guarantee agreement with regulatory targets.
3. Interpretability is regime-dependent: alignment between interpretability layers improves in cleaner, directly regulated settings but remains incomplete.
4. Interpretability is perturbation-specific: in heterogeneous datasets, mechanistic support varies across perturbations and cannot be inferred from regulator class

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5. Methods

5.1 Model architecture and concept prior

We used a concept-bottleneck neural network (MM-KPNN) in which gene expression inputs are mapped to an intermediate layer representing transcription factor (TF) concepts, followed by a task-specific output layer.

Let $x \in \mathbb{R}^G$ denote the input gene-expression vector for G genes. The intermediate representation consists of C predefined concepts corresponding to transcription factor regulons and related biological programs. The gene-to-concept mapping is implemented using a learnable weight matrix $W \in \mathbb{R}^{G \times C}$, constrained by a binary gene–concept mask $M \in \{0, 1\}^{G \times C}$ through elementwise multiplication:

$$W_{\text{masked}} = W \odot M$$

Concept pre-activations are computed as:

$$z_{\text{pre}} = xW_{\text{masked}} + b_c$$

where $b_c \in \mathbb{R}^C$ is a learned concept bias. Concept activations are obtained via an elementwise nonlinearity:

$$z = \phi(z_{\text{pre}})$$

where ϕ is ReLU in the primary analyses.

The output layer is a dense linear mapping from concepts to task logits:

$$\hat{y} = zW_o + b_o$$

Concepts are defined through a predefined concept library constructed prior to model training. The library is provided in Gene Matrix Transposed (GMT) format, where each concept corresponds to a set of associated genes. Transcription factor regulons were derived from curated regulatory resources, including DoRothEA (Garcia-Alonso et al., 2019), which provides experimentally supported TF–target interactions with associated confidence levels. Additional immune-related gene programs were included to capture broader biological states relevant to the datasets analyzed.

During preprocessing, the GMT file is converted into a binary gene–concept matrix M, which defines the connectivity of the gene-to-concept layer and is fixed throughout training. The same concept library is used across all datasets, ensuring consistency of the concept space. Model training was performed end-to-end using cross-entropy loss for classification tasks and the Adam optimizer. Training was implemented in PyTorch (Paszke et al., 2019), and preprocessing and data handling were performed using Scanpy (Wolf et al., 2018). In PBMC and Norman analyses, training was performed on full tensors, whereas in the Replogle analysis mini-batch training was used via PyTorch DataLoaders..

5.2 Datasets

We evaluated the model on datasets representing distinct biological regimes:

- PBMC stimulation dataset: peripheral blood mononuclear cells subjected to defined stimuli (Control, IFN, LPS, PMA), representing stimulus-driven transcriptional responses (Kartha et al., 2022).
- IFN-focused dataset: interferon-driven responses providing a more directly regulated setting (Kang et al., 2018).
- Norman Perturb-seq dataset: CRISPR perturbations targeting regulatory genes, producing distributed and indirect transcriptional effects (Norman et al., 2019).
- Replogle CRISPR dataset: a large-scale perturbation dataset spanning transcription factors, chromatin regulators, and other regulatory proteins (Replogle et al., 2022).
- Additional analyses were performed on a heterogeneous multicellular melanoma dataset to assess robustness in a complex biological setting (Tirosh et al., 2016).

These datasets were selected to represent complementary regimes of biological variation, enabling evaluation across stimulus-driven, directly regulated, indirect perturbation, and heterogeneous perturbation settings.

5.3 Concept activations

For each sample or perturbation, the model produces a vector of concept activations z, representing inferred activity of transcription factor programs.

Concept activation matrices were aggregated at the condition or perturbation level by averaging across cells or samples within each group. For the PBMC dataset, activation differences were computed relative to the control condition to obtain stimulus-specific activation shifts.

5.4 Concept selectivity

Concept selectivity quantifies the degree to which activation is concentrated in a single concept. For a given activation vector z, selectivity is defined as:

$$\textit{selectivity} = \max(z) - \textit{mean}(z \setminus \max(z))$$

This measure captures how strongly a single concept dominates the representation relative to the remaining concepts. Because selectivity values depend on dataset-specific scaling, comparisons across datasets are interpreted qualitatively rather than as directly normalized quantitative measures.

5.5 Concept-space geometry

Concept-space geometry refers to the organization of samples or perturbations in the space of concept activations. Principal component analysis (PCA) was applied to concept activation matrices to obtain low-dimensional embeddings.

The first two principal components were used to visualize clustering, separation, and global structure of perturbation responses. PCA was performed on centered concept activation matrices without additional scaling.

5.6 Gene-level mechanistic support

Gene-level support quantifies agreement between model-derived gene attributions and expected regulatory targets. Two complementary attribution strategies were used:

(i) Concept-level support: genes were ranked based on the learned masked gene-to-concept weights Wmasked for each transcription factor concept.

(ii) Perturbation-level support: gene importance was computed using input × gradient with respect to the true class logit, capturing the contribution of each gene to the model prediction for a given perturbation (Lundberg and Lee, 2017). For each case, the top-k genes (with fixed k)

were selected and compared to expected target sets. Overlap was computed as:

$$overlap = |\, top\text{-}k \cap expected\ targets \,| \,/\, k$$

Overlap is interpreted relative to a shuffled baseline obtained by randomly permuting gene labels and recomputing overlap. This provides a reference for random expectation and allows assessment of enrichment beyond chance. Expected target sets are derived from predefined regulatory annotations associated with each transcription factor. Overlap therefore provides an approximate measure of mechanistic consistency rather than a definitive indicator of causal regulation.

5.7 Classification of interpretability regimes

To summarize patterns across datasets, concept responses were categorized into qualitative interpretability classes based on activation and gene-level support:

- Strong mechanistic: high activation and high overlap
- Moderate mechanistic: intermediate activation and/or overlap
- Weak or non-mechanistic: low overlap regardless of activation

These categories are descriptive and are not intended as strict statistical thresholds.

5.8 Stability analysis

Stability of concept representations was evaluated across multiple model runs. For each run, concept activations, selectivity values, and gene-level support metrics were recomputed.

Stability was assessed by comparing:

- consistency of concept-space geometry
- variability in dominant concept identities
- variability in gene-level attributions

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Figures

Figure 1. Multi-layer interpretability framework for concept-bottleneck models
Gene expression inputs are mapped to transcription factor (TF) concepts through a concept bottleneck, followed by task-specific prediction. Interpretability is evaluated along three complementary components applied to the same learned representation: (i) concept selectivity, quantifying how concentrated activity is within individual concepts; (ii) concept-space geometry, describing the global organization of samples or perturbations; and (iii) gene-level mechanistic support, assessing agreement between attributed genes and expected regulatory targets. These components provide independent views of interpretability and do not imply a hierarchy.

Figure 2. PBMC: stimulus-specific concept activation and partial mechanistic alignment
(A) Concept activation across stimulation conditions (Control, IFN, LPS, PMA) shows stimulus-specific responses, including strong induction of interferon-related concepts (e.g., STAT2, IRF9, CIITA).
(B) Activation shifts relative to control highlight stimulus-dependent changes in concept activity.
(C) Relationship between activation shift and gene-level support (top-k overlap with expected targets) shows that some concepts exhibit both strong activation and strong support, while others display

weaker or inconsistent agreement. This defines a regime of partial mechanistic alignment.

Figure 3. Norman: structured concept-space geometry without gene-level support
(A) Perturbations form structured patterns in concept space (PCA projection), indicating organized responses at the representation level despite indirect transcriptional effects.
(B) Gene-level support is indistinguishable from a shuffled baseline, indicating no measurable enrichment of expected targets.
(C) Concept selectivity does not predict gene-level support. Together, these results demonstrate that structured concept-space organization can arise independently of gene-level interpretability.

Figure 4. Replogle: heterogeneous gene-level support without class-level predictability
(A) Gene-level support varies substantially within regulator classes (transcription factors, chromatin regulators, others), indicating high intra-class variability.
(B) Concept selectivity spans a wide range within each class, with no clear separation between regulator categories.
(C) Perturbation-level variation dominates class-level trends, showing that gene-level support is heterogeneous and cannot be inferred from regulator class.

Figure 5. Cross-dataset synthesis: regime-dependent gene-level support across structured representations
(A) Across datasets, concept selectivity does not consistently predict gene-level support.
(B) Median gene-level support differs by dataset, with higher values in stimulus-driven and IFN settings compared to perturbation datasets.
(C) Summary of observed regimes: all datasets exhibit structured concept-space organization, but gene-level support varies—partial in PBMC, strongest in IFN, absent in Norman, and heterogeneous in Replogle.

Supplementary Information

Reproducible analyses are provided as Jupyter notebooks accompanying this study. Each notebook corresponds to a dataset-specific or cross-dataset analysis and can be executed sequentially to reproduce the main results. All key outputs are displayed directly within the notebooks and annotated to facilitate interpretation.

The supplementary materials include:

Supplementary Notebook 1 — PBMC analysis
Analysis of concept activation patterns under stimulus conditions, including activation shifts relative to control and evaluation of gene-level mechanistic support.

Supplementary Notebook 2 — Norman Perturb-seq analysis
Evaluation of concept-space geometry in a perturbation-driven setting, including PCA-based visualization of structured responses.

Supplementary Notebook 3 — Replogle perturbation analysis
Assessment of gene-level support across perturbations, highlighting heterogeneity and lack of class-

level predictability.

Supplementary Notebook 4 — Cross-dataset analysis
Comparative analysis across PBMC, IFN, Norman, and Replogle datasets, summarizing regime-dependent differences in interpretability.

Supplementary Notebook 5 — Figure generation pipeline
End-to-end pipeline for reproducing all main figures, including preprocessing, model evaluation, and visualization steps.

Figure 1. Framework: multi-layered interpretability

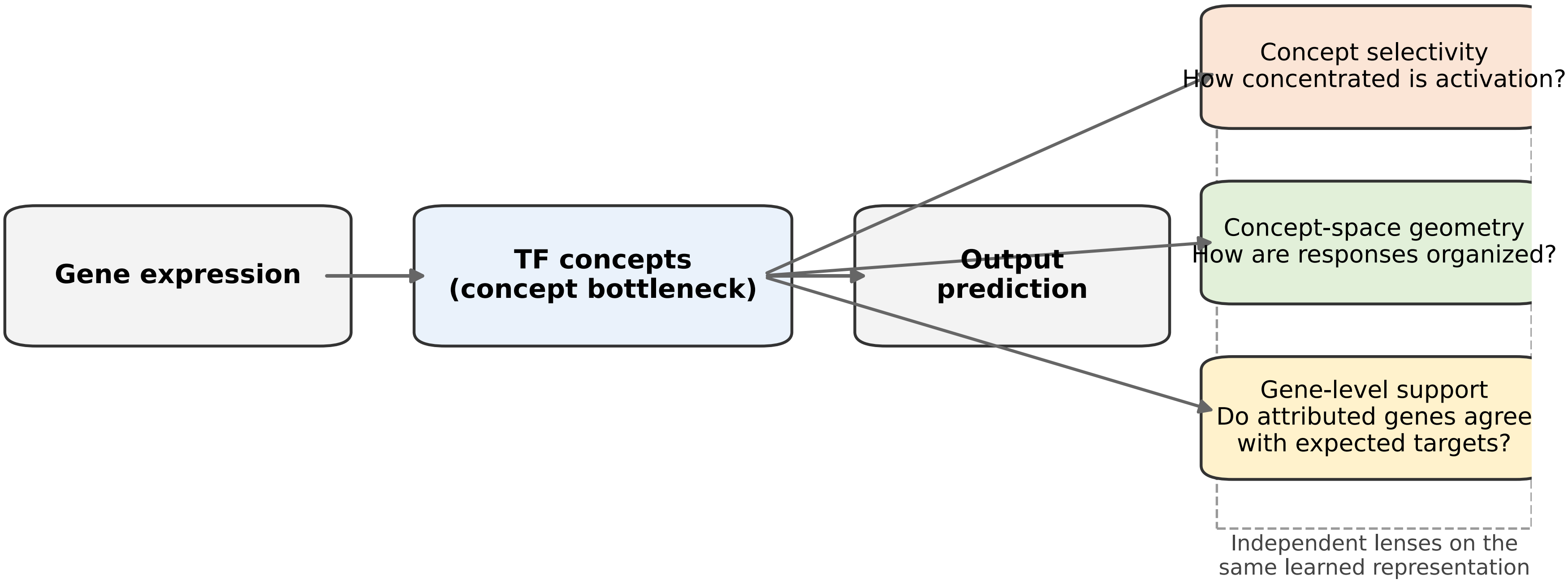


Figure 2. PBMC: partial alignment of concept activity and gene-level support

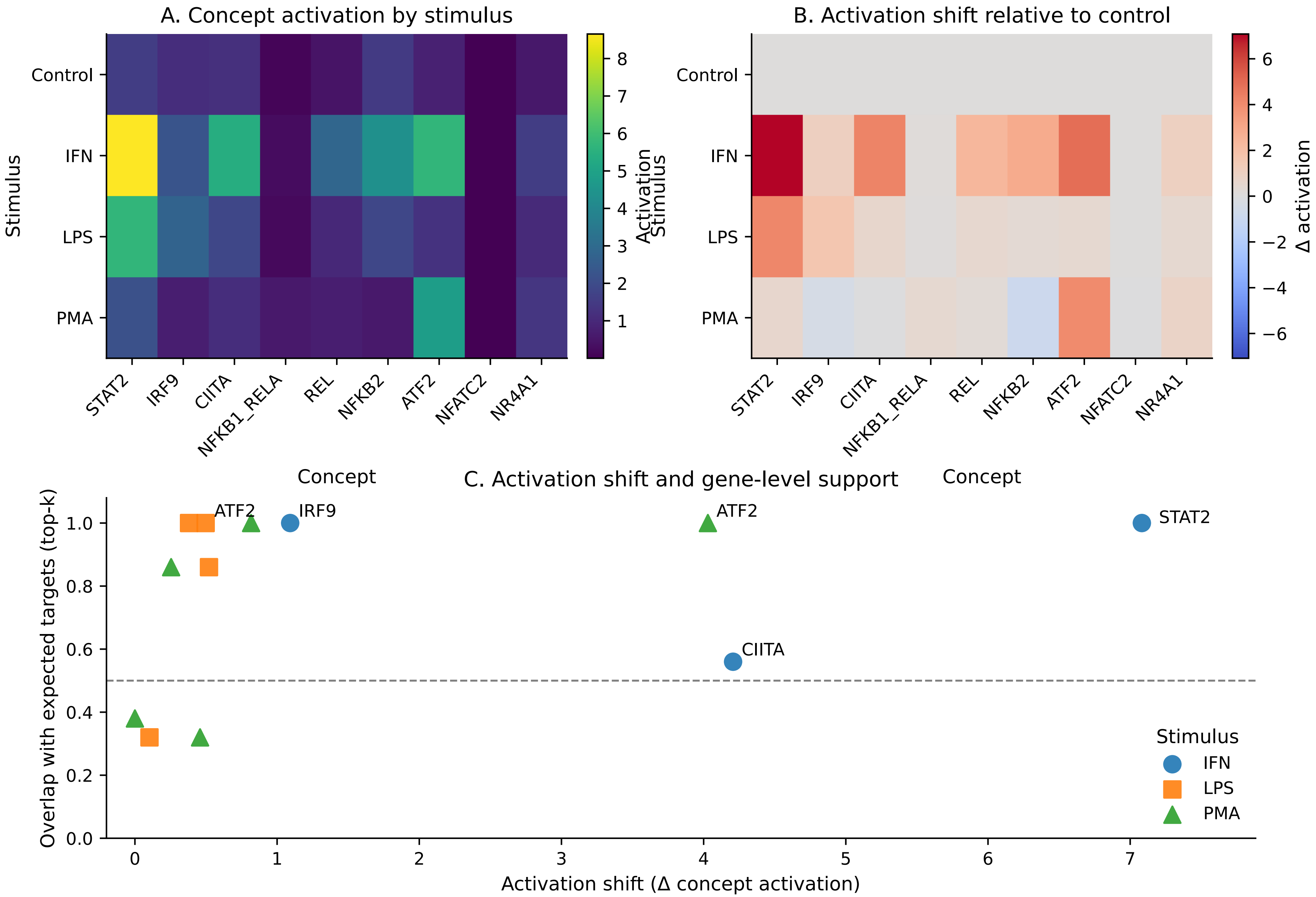


Figure 3. Norman: structured concept-space responses without gene-level support

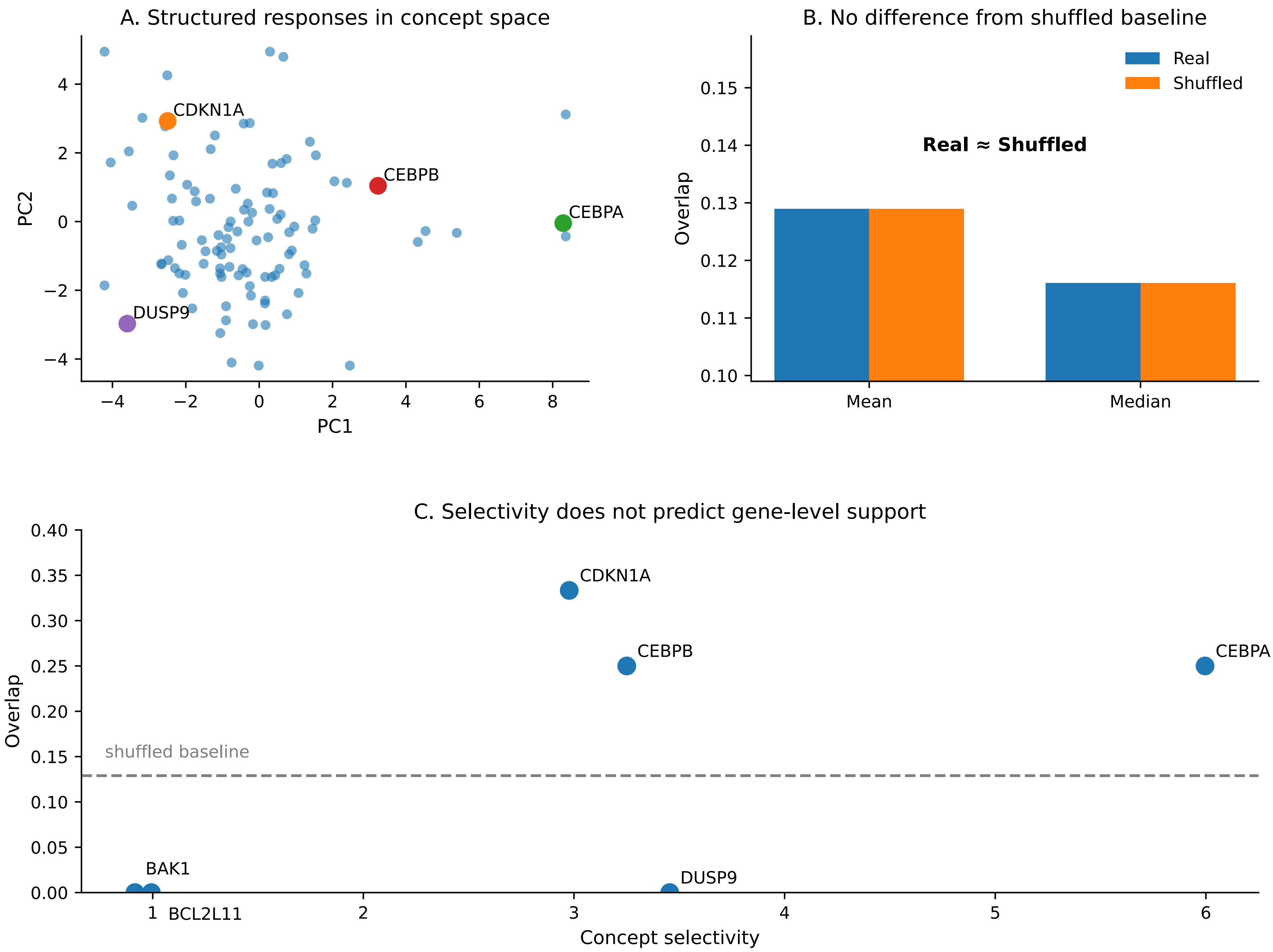


Figure 4. Replogle: heterogeneous gene-level support without clear class-level separation

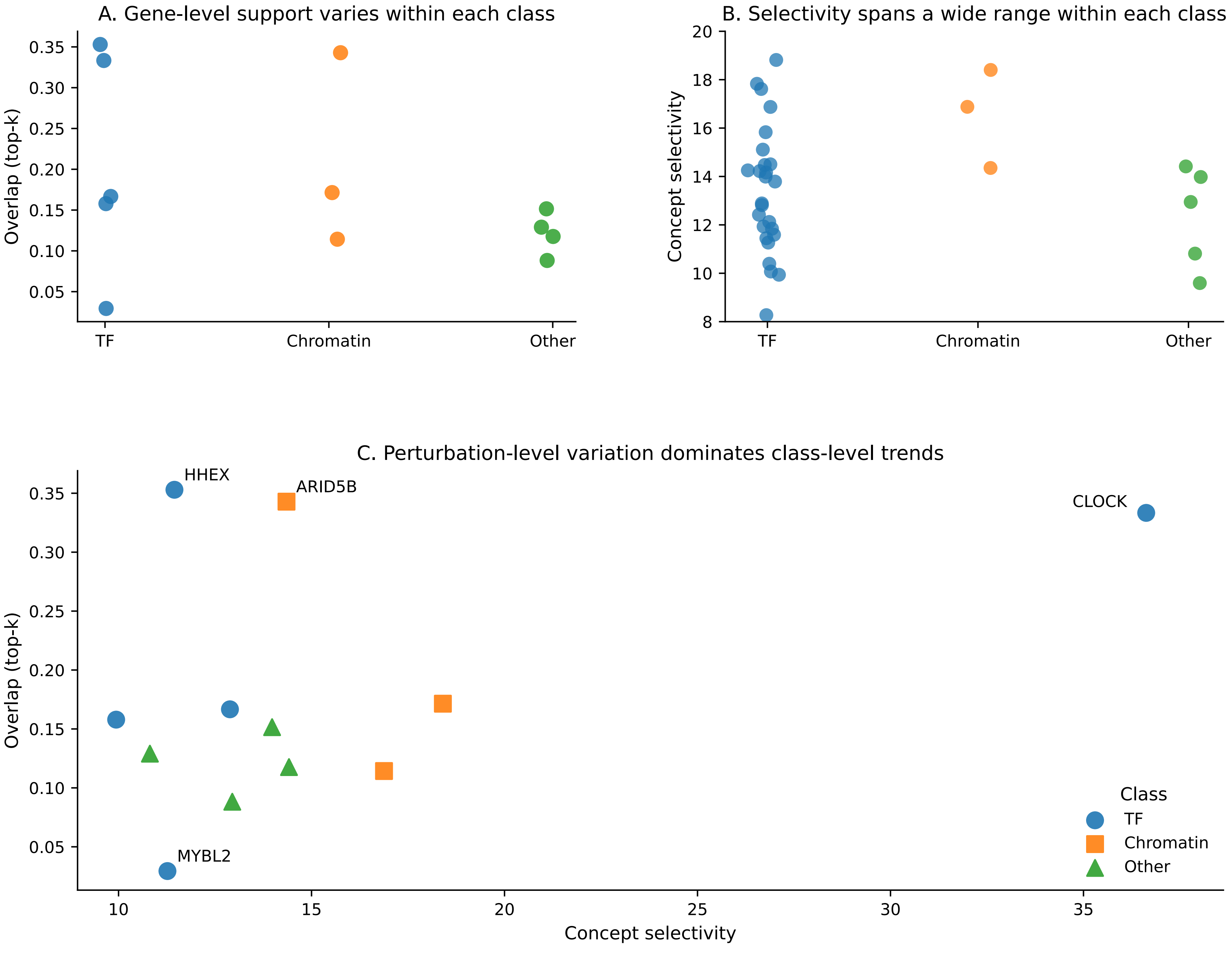
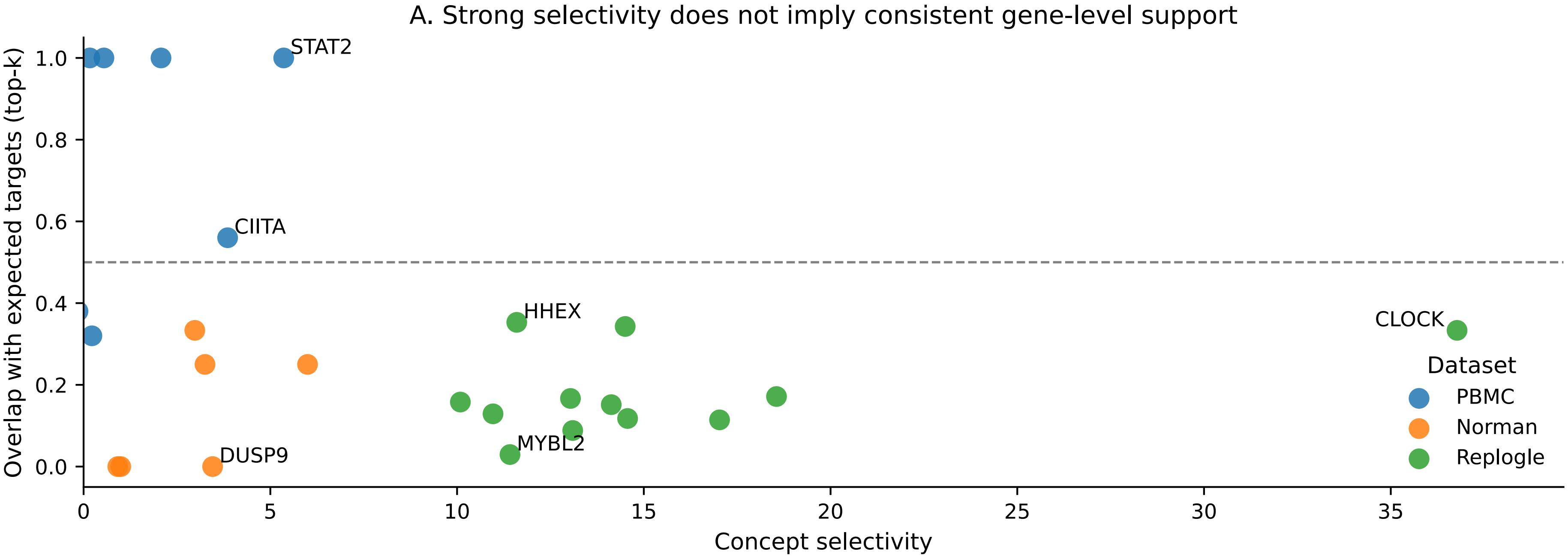
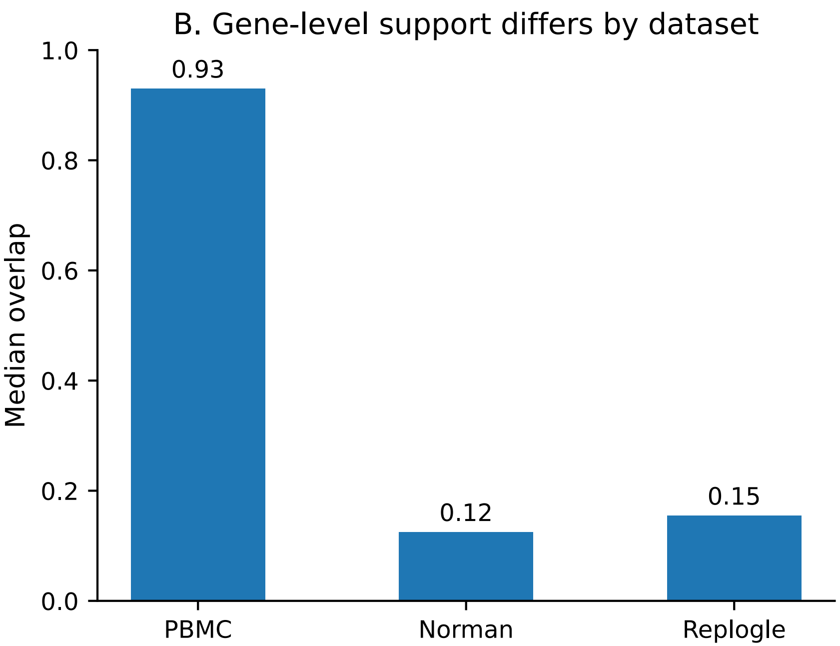


Figure 5. Cross-dataset synthesis: regime-dependent gene-level support across structured responses





C. Interpretive summary

Dataset	Representation	Gene-level support
PBMC	Structured (Fig. 2)	Partial / subset-dependent
Norman	Structured (Fig. 3)	Weak
Replogle	Structured (Fig. 4)	Heterogeneous